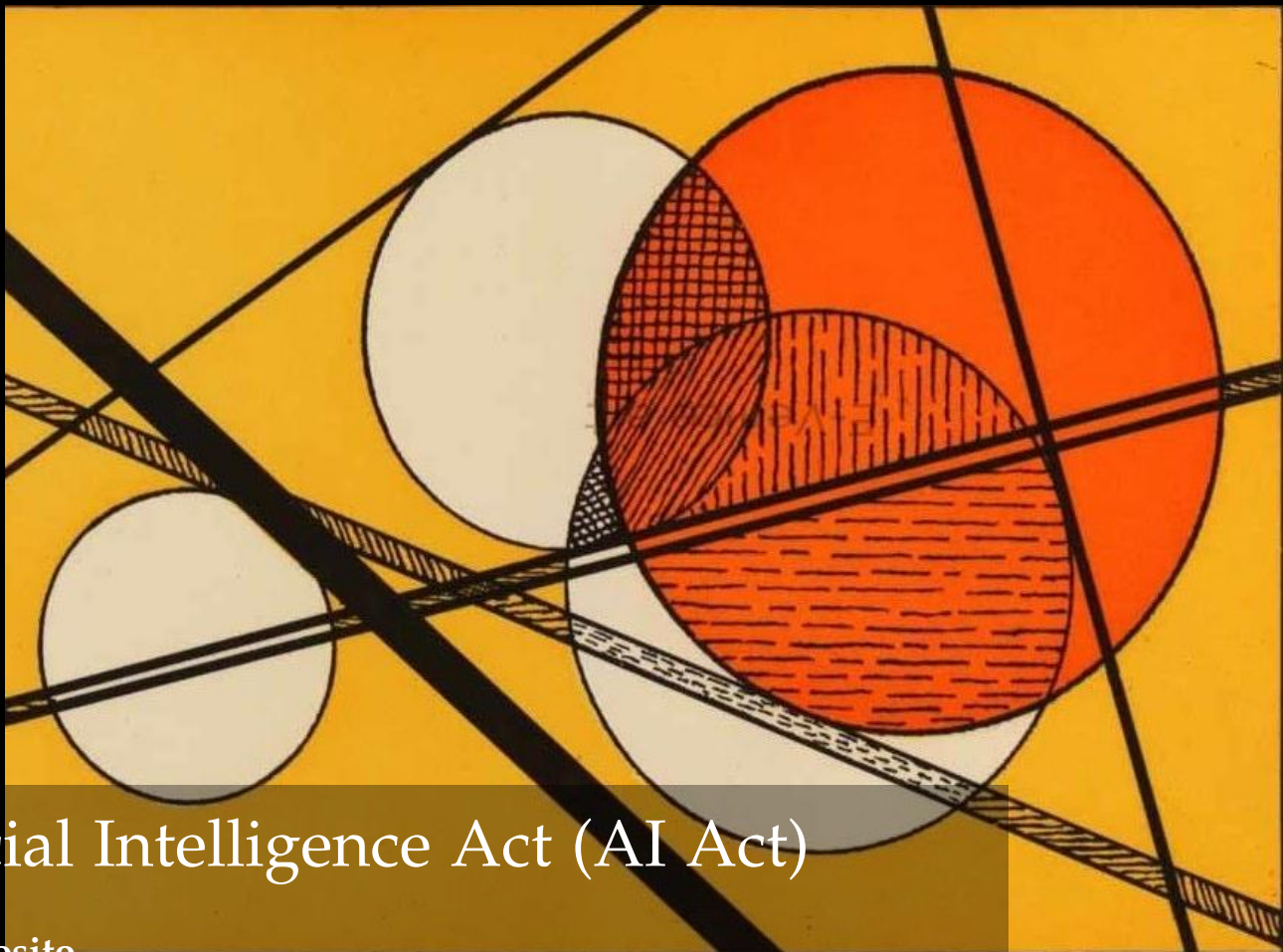


Research Paper Series



Artificial Intelligence Act (AI Act)

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Executive Summary

The purpose of *Artificial Intelligence Act* (AI Act) is to improve the functioning of the internal European market by laying down a uniform legal framework in particular for the development, the placing on the market, the putting into service and the use of AI systems to promote the uptake of human centric and trustworthy AI while ensuring a high level of protection of health, safety, fundamental rights as enshrined in the Charter of fundamental rights of the European Union. The Act seeks to balance the drive for technological innovation and emphasizes the need to prevent the fragmentation of the internal market due to divergent national regulations, thus facilitating the free movement and cross-border deployment of AI-based goods and services.



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Artificial Intelligence Act (AI Act)

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ARTIFICIAL Intelligence (AI) is a fast-evolving family of technologies that contributes to a wide array of economic, environmental and societal benefits across the entire spectrum of industries and social activities. By improving prediction, optimising operations and resource allocation, and personalising digital solutions available for individuals and organisations, the use of AI can provide key competitive advantages to undertakings and support socially and environmentally beneficial outcomes. At the same time, depending on the circumstances regarding its specific application, use, and level of technological development, AI may generate risks and cause harm to public interests and fundamental rights that are protected by European Union law.

AI systems can be easily deployed in a large variety of sectors of the economy. Certain Member States have already explored the adoption of national rules to ensure that AI is trustworthy and safe and is developed and used in accordance with fundamental rights obligations. Diverging national rules may lead to the fragmentation of the internal market and may decrease legal certainty for operators that develop, import or use AI systems. With the Artificial Intelligence Act (AI Act) Regulation, however, a high and consistent level of protection is guaranteed throughout the European Union can therefore be ensured in order to achieve trustworthy AI, while divergences hampering the free circulation, innovation, deployment and the uptake of AI systems and related products and services within the internal market can be prevented by laying down uniform obligations for operators and guaranteeing the uniform protection of overriding reasons of public interest and of rights of persons throughout the internal market.

In particular, in the banking sector the use of AI by operators is generating a series of changes that will have an impact, sometimes radical, on various profiles: on customers through a modification of their purchasing experience; on the productivity of operators through a change in production processes and the value chain; for the authorities, following the review of risk measurement and management techniques and the implications for supervisory tools. As regards use cases, there are many applications in the banking sector: from marketing to combating fraud in transaction banking and payments, from scoring models for credit risk to risk management systems[2].

1. The Artificial Intelligence Act (AI Act)

On 21 April 2021[9], the Commission presented a proposal for a regulation aimed at harmonizing the rules on artificial intelligence (AI Regulation) and a coordinated plan comprising a series of joint actions for the Commission and Member States. This package of rules aimed to increase trust in AI and promote the development and upgrading of AI technologies. On December 2022[4], the European Council has adopted its common position (*"general approach"*) on the AI Act and in June 2023[12], the European Parliament has adopted their negotiation position for the draft AI Act. On 9 December 2023[10], the Council and Parliament reach a provisional agreement on the AI Regulation after months of negotiations. **During 2024, both the Council (21 May 2024[11]) and the Parliament (13 March 2024[13]), as EU co-legislators, formally adopted the final text.** The AI Act will therefore enter into force on the twentieth day following its publication in the Official Journal of the EU and will be fully applicable after 24 months. However, some specific provisions will have different application dates, such as bans relating to AI systems with an unacceptable level of risk, which will

be applicable starting 6 months after entry into force, or provisions relating to GPAI models already in place on the market, for which a deadline of 12 months is envisaged to guarantee compliance with the provisions of the Regulation (for further details please refer to the chapter "Entry into force and review").

The purpose of the AI Act Regulation is to improve the functioning of the internal market and promote the uptake of human-centric and trustworthy artificial intelligence (AI), while ensuring a high level of protection of health, safety, fundamental rights enshrined in the Charter of Fundamental Rights, including democracy, the rule of law and environmental protection, against the harmful effects of artificial intelligence systems (AI systems) in the European Union, and to support innovation. To achieve that objective, rules regulating the placing on the market, the putting into service and the use of certain AI systems was laid down, thus ensuring the smooth functioning of the internal market and allowing those systems to benefit from the principle of free movement of goods and services. Those rules are clear and robust in protecting fundamental rights, supportive of new innovative solutions, enabling a European ecosystem of public and private actors creating AI systems in line with EU values and unlocking the potential of the digital transformation across all regions of the EU. By laying down those rules as well as measures in support of innovation with a particular focus on small and medium enterprises (SMEs), including startups, this Regulation supports the objective of promoting the European human-centric approach to AI and being a global leader in the development of secure, trustworthy and ethical AI as stated by the European Council. In the AI Act, an "AI system" is considered a machine-based system designed to operate with varying levels of autonomy, that may exhibit adaptiveness after deployment and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. This notion of "AI system" is closely aligned with the work of international organisations working on AI to ensure legal certainty, facilitate international convergence and wide acceptance, while providing the flexibility to accommodate the rapid technological developments in this field. Moreover, it is based on key characteristics of AI systems that distinguish it from simpler traditional software systems or programming approaches and not cover systems that are based on the rules defined solely by natural persons to automatically execute operations.

The techniques that enable inference while building an AI system include machine learning approaches that learn from data how to achieve certain objectives, and logic - and knowledge-based approaches that infer from encoded knowledge or symbolic representation of the task to be solved. The capacity of an AI system to infer transcends basic data processing, enables learning, reasoning or modelling.

In order to obtain the greatest benefits from AI systems while protecting fundamental rights, health and safety and to enable democratic control, AI literacy¹ must equip providers, deployers and affected persons with the necessary notions to make informed decisions regarding AI systems. AI literacy must provide all relevant actors in the AI value chain with the insights required to ensure the appropriate compliance and its correct enforcement.

Also, in order to ensure a level playing field and an effective protection of rights and freedoms of individuals across the European Union, AI Act it also recognizes the importance of international cooperation, aiming for regulatory consistency and fostering global dialogue on ethical AI standards. The regulation applies in a non-discriminatory manner broadly to AI systems' providers, deployers, importers, distributors, and manufacturers within the EU and those from third countries whose AI outputs are used in the EU. Furthermore, it explicitly includes provisions for authorized representatives and affected persons in the European Union.

The introduction of AI Act previews the Act's innovations, including measures to support AI innovation, particularly for SMEs, and the establishment of an EU database for high-risk AI systems. It also outlines proposed amendments to integrate the Act's requirements with existing EU legislation, ensuring a cohesive and comprehensive regulatory landscape.

As already reported above, AI can bring numerous benefits for society and the economy but, at the same time, it can also present risks for rights, security and the proper functioning of the single market. To find a balance between the two instances, the AI Act was designed with a risk-based

¹"AI literacy" means skills, knowledge and understanding that allows providers, deployers and affected persons, taking into account their respective rights and obligations in the context of this Regulation, to make an informed deployment of AI systems, as well as to gain awareness about the opportunities and risks of AI and possible harm it can cause.

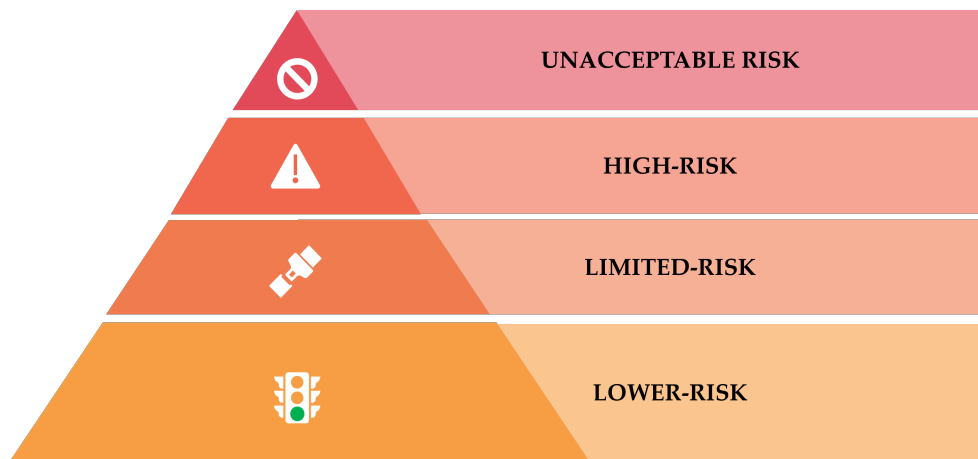


FIGURE 1: AI Act risk-based approach

approach. The regulatory framework on AI therefore provides for a classification of AI systems based on the level of risk they present for people and society. The framework distinguishes between four categories of risk: unacceptable, high, limited and minimal (as shown in the Figure 1):

- AI systems that pose an **unacceptable risk** are those that contradict fundamental EU values and principles, such as respect for human dignity, democracy and the rule of law. These systems are banned or (in the case of real-time biometric surveillance for security reasons) subject to severe restrictions. For example, AI systems that manipulate human behavior in a way that circumvents the will of users, or that enable "social scoring"² by public authorities are prohibited.
- AI systems that present a **high risk** are those that can have a "systemic" impact, i.e. a significant impact on the fundamental rights or safety of people. These systems are subject to rigorous obligations and requirements before they can be placed on the market or used. For example, this category includes AI systems used to evaluate the credit score or creditworthiness³ of natural persons and the selection and recruitment of personnel, for admission to education, for the provision of essential social services, such as healthcare, for remote biometric surveillance (not in real time), for judicial and police applications, or for the management of critical infrastructure security.
- AI systems that present **limited risk** are those that can influence the rights or wishes of users, but to a lesser extent than high-risk systems. These systems are subject to transparency requirements, which allow users to be aware that they are interacting with an AI system and to understand its characteristics and limitations. For example, AI systems used to generate or manipulate audiovisual content (such as deepfakes), or to provide personalized suggestions (such as chatbots) fall into this category. There is a right to know that you are talking to a bot (instead of a human) and that that image is created or contrived by AI.
- AI systems that pose **minimal risk** are those that have no direct impact on fundamental rights or the safety of people, and that offer wide margins of choice and control to users. These systems are free from any regulatory obligations, in order to encourage innovation

²Systems aimed to classify or evaluate people or groups of people 'based on their social behavior or known or predicted personal or personality characteristics' resulting in a "social score" to the detriment of these people. These AI-based practices open or restrict access to social benefits and/or differentiate treatment based on a score obtained from assessing personal behaviors or attributes.

³AI systems used to evaluate the credit score or creditworthiness of natural persons should be classified as high-risk AI systems; AI systems used for those purposes may lead to discrimination between persons or groups and may perpetuate historical patterns of discrimination, such as that based on racial or ethnic origins, gender, disabilities, age or sexual orientation, or may create new forms of discriminatory impacts. However, AI systems provided for by Union law for the purpose of detecting fraud in the offering of financial services and for prudential purposes to calculate credit institutions' and insurance undertakings' capital requirements should not be considered to be high-risk under this Regulation.

and experimentation. For example, AI systems used for recreational purposes (such as video games) or for purely aesthetic purposes (such as photographic filters) fall into this category.

Among others, the following systems are excluded from the obligations of the AI Act Regulation:

- AI systems used exclusively for military, defense, or national security purposes;
- Public authorities and international organizations using AI in law enforcement and judicial cooperation, provided there are adequate safeguards for fundamental rights;
- AI systems developed solely for scientific research;
- AI systems or models before their market introduction or service provision;
- Personal, non-professional use of AI systems by natural persons.

Even through the risk-based approach is the basis for a proportionate and effective set of binding rules, it is important to recall the “2019 Ethics guidelines for trustworthy AI”[8] developed by the High-Level Expert Group (HLEG) on AI. In those guidelines, the AI HLEG developed seven non-binding ethical principles for AI which are intended to help ensure that AI is trustworthy and ethically sound. The seven principles include human agency and oversight; technical robustness and safety; privacy and data governance; transparency; diversity, non-discrimination and fairness; societal and environmental well-being and accountability. Those guidelines contribute to the design of a coherent, trustworthy and human-centric AI, in line with the Charter and with the values on which the European Union is founded.

In the continuation of the examination of this paper the focus is almost exclusively on the first two risk categories: *Prohibited AI practices* and *High risk AI system*.

1.1 Prohibited AI Practices

As already indicated before, AI Act outlines specific AI practices that are deemed unacceptable and are thus prohibited under the Act; this represents a crucial component of the EU’s regulatory approach towards ensuring that the development and application of AI technologies align with fundamental rights, ethical standards, and societal values. The prohibited practices target AI systems that could potentially harm individuals’ autonomy, safety, privacy, and rights, emphasizing the EU’s commitment to a human-centric AI approach.

Primarily, the regulation explicitly bans AI systems that deploy **subliminal, manipulative, or deceptive techniques capable of materially distorting a person’s behavior by impairing their ability to make informed decisions**. This prohibition targets AI systems that, through manipulation beyond a person’s consciousness, could cause significant harm by leading individuals to make choices they would not have otherwise made. The underlying concern here is the protection of individual autonomy and the prevention of exploitation through AI-driven manipulation.

Secondly, AI systems designed to **exploit the vulnerabilities of specific groups, including those defined by age, disability, or socio-economic status, are prohibited**. This provision aims to prevent AI from materially distorting the behavior of vulnerable individuals in a way that could cause significant harm. It reflects the EU’s commitment to safeguarding the rights and well-being of all citizens.

In addition to the practices already listed, the Act prohibits AI systems that evaluate or classify groups or individuals based on their social behavior or predicted personal characteristics over time, leading to detrimental or unfavorable treatment. **Social scoring** systems, which could result in unfair, disproportionate, or unrelated consequences in various social contexts, are seen as antithetical to EU values, including respect for human dignity and equality before the law.

Moreover, prohibited are AI systems used for making risk assessments of individuals to **predict their likelihood of committing a crime based solely on profiling or personality traits**. This prohibition, however, does not extend to AI systems supporting human assessments in criminal investigations where the analysis is based on objective facts directly linked to criminal activity. The focus here is on preventing unjust or biased law enforcement practices that could undermine fundamental rights, such as the presumption of innocence.

The Act bans also the use of AI for untargeted scraping of **facial images to create or expand facial**

recognition databases. This prohibition addresses privacy concerns related to mass surveillance and the unauthorized collection of biometric data, emphasizing the importance of consent and the protection of personal data. In addition to what has already been reported above, prohibited is also the use of AI to **infer emotions in contexts such as the workplace and educational institutions**, except for medical or safety reasons. This prohibition aims to protect individuals from invasive assessments that could lead to discrimination or adverse consequences based on inferred emotional states.

Furthermore, the regulation bans AI systems that **categorize individuals based on biometric data to infer sensitive attributes**, such as race, political opinions, trade union membership, religious or philosophical beliefs, sex life or sexual orientation. This prohibition does not cover any labelling or filtering of lawfully acquired biometric datasets, such as images, based on biometric data or categorizing of biometric data in the area of law enforcement. This prohibition seeks to prevent discriminatory practices and protect individuals' privacy and fundamental rights, particularly in sensitive areas.

Finally, the use of **real-time remote biometric identification systems** by law enforcement in publicly accessible spaces is forbidden, unless and in so far as such use is strictly necessary for one of the following objectives:

- The targeted search for specific victims of abduction, trafficking in human beings or sexual exploitation of human beings, as well as searching for missing persons.
- The prevention of a specific, substantial and imminent threat to the life or physical safety of natural persons or a genuine and present or genuine and foreseeable threat of a terrorist attack.
- The localisation or identification of a person suspected of having committed a criminal offence, for the purpose of conducting a criminal investigation, prosecution or executing a criminal penalty.

This provision balances the need for public safety with the protection of privacy and fundamental freedoms, establishing stringent conditions for the deployment of such technologies.

Detailed conditions under which the limited exceptions for the use of real-time remote biometric identification systems may be applied are also established. These include the requirement for a fundamental rights impact assessment, registration of the system in the EU database, and strict temporal, geographic, and personal limitations to ensure that the use is necessary, proportionate, and respectful of individuals' rights and freedoms. The deployment of real-time remote biometric identification systems must be preceded by authorization from a judicial or independent administrative authority, ensuring oversight and accountability. In urgent situations, temporary use without prior authorization is permitted, provided that authorization is sought immediately thereafter. Member States have the discretion to authorize the use of real-time remote biometric identification systems within the framework set by the Act, reflecting the principle of subsidiarity and allowing for national variations in implementation. Each use of real-time remote biometric identification systems must be notified to relevant authorities, including market surveillance and national data protection authorities and Member States are required to report annually on the use of such systems; the Commission is tasked with publishing aggregated reports, ensuring transparency and public accountability.

1.2 High-Risk AI System

The regulatory framework for the classification, assessment, and management of AI systems deemed to be of high risk aims to ensure that **high-risk AI systems** are developed, deployed, and used in a manner that respects fundamental rights, ensures safety, and fosters trust among users and the broader public. In the AI Act, an *"high-risk"* is considered when the following conditions are fulfilled:

- The AI system is intended to be used as a safety component;
- The AI system is required to undergo a third-party conformity assessment, with a view to the placing on the market or the putting into service of that product;

- The AI system is always considered to be high-risk where performs profiling of natural persons.

Otherwise, an AI system shall not be considered to be high-risk if it does not pose a significant risk of harm to the health, safety, or fundamental rights of natural persons, including by not materially influencing the outcome of decision making. This shall be the case where it performs a narrow procedural task, improves the result of a previously completed human activity, detects decision-making patterns or deviations from prior decision-making patterns and is not meant to replace or influence the previously completed human assessment, without proper human review and carrying out a preparatory activity for a purpose-relevant evaluation.

The use-cases of high-risk AI systems could be modified from European Commission when its purposes pose a risk of harm to health and safety, or an adverse impact on fundamental rights. When assessing the condition, the classification rules take into account the following criteria:

- The intended purpose;
- How much is involved within the process;
- The nature - i.e., the presence of special categories of personal data are processed - and amount of the data processed and used by the AI system;
- How autonomous the system acts and the possibility for a human to override a decision or recommendations that may lead to potential harm;
- The extent to which the use of an AI system has already caused harm to health and safety, has had an adverse impact on fundamental rights or has given rise to significant concerns in relation to the likelihood of such harm or adverse impact;
- The intensity of such harm or such adverse impact on a single or a multiple person;
- How important is the result achieved by the AI system despite the negative impact;
- How much is the imbalance between those negatively affected by the AI system and those who employ it;
- The extent to which the outcome produced involving an AI system is easily corrigible or reversible, taking into account the technical solutions available to correct or reverse it, whereby outcomes having an adverse impact on health, safety or fundamental rights, shall not be considered to be easily corrigible or reversible;
- The magnitude and likelihood of benefit of the deployment of the AI system for individuals, groups, or society at large, including possible improvements in product safety;
- The extent to which existing EU law provides for:
 - Effective measures of redress in relation to the risks posed by an AI system, with the exclusion of claims for damages;
 - Effective measures to prevent or substantially minimise those risks.

Finally, the European Commission is granted the power to removing high-risk AI systems where both of the following conditions are fulfilled:

- The high-risk AI system concerned no longer poses any significant risks to fundamental rights, health or safety;
- The deletion does not decrease the overall level of protection of health, safety and fundamental rights under EU law.

High-risk AI systems must comply with specified requirements, considering their intended purposes and the state of the art in AI technologies. Providers are responsible for ensuring full compliance with applicable EU legislation, are responsible for high-risk AI systems and guarantee consistency, avoid duplications and minimise additional burdens. They also shall have a choice of integrating,

as appropriate, the necessary testing and reporting processes, information, and documentation regarding their product. In order to be able to represent all the information mentioned above, it is necessary to have a continuous iterative process planned and managed throughout the entire lifecycle of the high-risk AI system. This document is called **risk management system**, is mandatory and requires regular systematic review and updating. In order to better intervene in its risk management, measurement and mitigation activities, it uses risk management measures that are effective in making the residual risk associated with each danger acceptable, as well as the overall residual risk of high-risk AI systems.

To this end it is necessary to have operators with adequate technical knowledge, as well as the experience, education and training expected and the ability to learn the context in which the system is intended to be used. Through the use of high-risk AI systems testing, it is possible to ensure that they function consistently, respecting the requirements and limits - previously defined parameters and adequate probabilistic thresholds - for their intended purpose. Moreover, high-risk AI systems which make use of techniques involving the training of AI models with data shall be developed on the basis of training, validation and testing data subject to data governance and management; in fact, they shall be relevant, sufficiently representative, and to the best extent possible, free of errors and complete in view of the intended purpose.

With regards to data, it is therefore important that correct data processing is carried out, all the following conditions apply:

- The bias detection and correction cannot be effectively fulfilled by processing other data, including synthetic or anonymised data;
- Regarding the special categories of personal data are subject to more stringent treatments:
 - There are technical limitations on the re-use of the personal data, and state of the art security and privacy-preserving measures, including pseudonymisation;
 - There are measures to ensure that the personal data processed are secured, protected, subject to suitable safeguards, including strict controls and documentation of the access, to avoid misuse and ensure that only authorised persons with appropriate confidentiality obligations have access to those personal data;
 - They are not to be transmitted, transferred or otherwise accessed by other parties;
 - They are deleted once the bias has been corrected or the personal data has reached the end of its retention period, whichever comes first;
- The records of processing activities include the reasons why the processing of special categories of personal data was strictly necessary to detect and correct biases, and why that objective could not be achieved by processing other data.

All this information shall be insert in a technical documentation shall be drawn up before that high-risk AI system is placed on the market or put into service. Everything reported within it must be continuously verified - in terms of compliance with the requirements provided by national competent authorities - and kept updated.

For specific categories, for example SME or start-up, the Commission shall establish a simplified technical documentation form targeted at the needs of small and microenterprises. Record-keeping ensures a level of traceability and facilitates the monitoring of the functioning of a high-risk AI system. The logging capabilities shall provide, at a minimum:

- Recording of the period of each use of the system (start date and time and end date and time of each use);
- The reference database against which input data has been checked by the system;
- The input data for which the search has led to a match;
- The identification of the natural persons involved in the verification of the results.

In addition to drawing up technical documentation, it is required that this has certain transparency characteristics. Transparency is essential to ensure compliance with the involved supplier and operator obligations. The instructions for use in an appropriate format that allows concise, complete, correct, and clear information as well as relevant, accessible and comprehensible to deployers. Technical documentation, record keeping and transparency are important factors that allow an effectively overseen by natural persons during the period in which they are in use.

High-risk AI systems shall be designed and developed in such a way to facilitate being overseen by natural persons during the period in which they are in use. Human oversight shall aim to prevent or minimise the risks to health, safety or fundamental rights that may emerge when the system itself is used improperly. The oversight measures shall be commensurate with the risks, level of autonomy and context of use of the high-risk AI system. Human oversight is assigned and enabled for those who are able to:

- Properly understand the relevant capacities and limitations of the high-risk AI system and be able to duly monitor its operation, including in view of detecting and addressing anomalies, dysfunctions, and unexpected performance;
- Recognize automatically relying or over-relying on the output produced by the system ("automation bias");
- Correctly interpret the high-risk AI system's output, considering, for example, the interpretation tools and methods available;
- Intervene in the operation through its interruption with a procedure that allows the system to come to a halt in a safe state.

High-risk AI systems shall be designed and developed in such a way that they achieve an appropriate level of accuracy, robustness, and cybersecurity, and that they perform consistently in those respects throughout their lifecycle. For achieve these levels of performance the Authorities, encourage, as appropriate, the development of benchmarks and measurement methodologies. The robustness of high-risk AI systems may be achieved through technical redundancy solutions (i.e., backup or fail-safe plans) and with mechanism resilient against attempts by unauthorised third parties to alter their use, outputs or performance by exploiting system vulnerabilities. Furthermore, continuous learning after being placed on the market or put into service shall be developed in such a way as to eliminate or reduce as far as possible the risk of possibly biased outputs influencing input for future operations ("*feedback loops*"), and as to ensure that any such feedback loops are duly addressed with appropriate mitigation measures. The technical solutions to address AI specific vulnerabilities shall include, in addition to cybersecurity, measures to prevent, detect, respond to, resolve and control for attacks trying to manipulate the training data set ("*data poisoning*"), or pre-trained components used in training ("*model poisoning*"), inputs designed to cause the AI model to make a mistake ("*adversarial examples*" or "*model evasion*"), confidentiality attacks or model flaws.

Concerning the responsibilities of parties involved in the lifecycle of high-risk AI systems the providers are subject to a range of obligations, including ensuring compliance with regulatory requirements, maintaining quality management systems. The tasks shall be kept in a comprehensive and systematic documentation and in an orderly manner in the form of written policies, procedures, and instructions. They shall ensure the degree of rigor and the level of protection required by the Regulation, conducting necessary corrective actions (to withdraw it, to disable it, or to recall it, as appropriate):

- **Providers** when are established in third countries shall, prior to making their high-risk AI systems available on the European Union market, appoint an authorised representative which is established in the EU. The authorised representative, by written mandate, perform the same tasks that each provider is required to do (verify and provide the technical documentation and all the information necessary to demonstrate the conformity of high-risk AI system and reduce and mitigate every possible risk involved). The authorised representative shall terminate the mandate if it considers or has reason to consider the provider to be acting contrary to its obligations pursuant to the AI Regulation and inform the market surveillance authority of the Member State about the termination of the mandate and the reasons therefor.

- **Deployers** that use high-risk AI systems shall take appropriate technical and organisational measures to ensure - according to provided instructions - the human oversight shall be assigned to natural persons who have the necessary competence, training and authority, as well as the necessary support. They shall exercise the control over the input data and monitor the operation of the high-risk AI system, and, when they identify a serious incident, they shall also immediately inform first the provider, and then the importer or distributor and the relevant market surveillance authorities of that incident. They are authorised to conserve the logs automatically generated of at least six months, unless provided otherwise in applicable EU or national law.

Before putting into service or using a high-risk AI system at the workplace, deployers who are employers shall inform workers' representatives and the affected workers that they will be subject to the use of the high-risk AI system. They also shall evaluate the high-risk AI systems' impact through assessment of the performs in terms of categories concerned (groups of person or specific risks), human oversight measures, internal governance, and complaint mechanisms; with the goal of taking the necessary steps to update the information and report any risks or incidents associated with the system's use.

Notifying authorities and notified bodies play a critical role in assessing and certifying the conformity of high-risk AI systems with regulatory requirements:

- **Notifying authorities:** each Member State shall designate or establish at least one notifying authority responsible for setting up and carrying out the necessary procedures for the assessment, designation, and notification of conformity assessment bodies and for their monitoring. Those shall be organised in such a way that:
 - No conflict of interest arises with conformity assessment bodies, and that the objectivity and impartiality of their activities are safeguarded;
 - Decisions relating to the notification of conformity assessment bodies are taken by competent persons different from those who carried out the assessment of those bodies;
 - Offer or provide neither any activities that conformity assessment bodies perform, nor any consultancy services on a commercial or competitive basis;
 - Have at their disposal an adequate number of competent personnel at their disposal for the proper performance of their tasks.
- **Notified bodies:** there are individual that depend on neither any the provider of a high-risk AI system in relation to which they perform conformity assessment activities, nor any other operator having an economic interest in high-risk AI systems assessed, as well as of any competitors of the provider. Those shall satisfy the organisational, quality management, resources and process requirements that are necessary to fulfil their tasks, as well as suitable cybersecurity requirements.

The organisational structure, allocation of responsibilities, reporting lines and operation of notified bodies shall ensure confidence in their performance, and in the results of the conformity assessment activities that the notified bodies conduct; in fact, they shall not engage in any activity that might conflict with their independence of judgement or integrity.

Notified bodies shall have documented procedures in place ensuring that their personnel, committees, subsidiaries, subcontractors and any associated body or personnel of external bodies maintain the confidentiality of the information which comes into their possession during the performance of conformity assessment activities, except when its disclosure is required by law.

Where a notified body finds that an AI system no longer meets the requirements taking account of the principle of proportionality, suspend or withdraw the certificate issued or impose restrictions on it, unless compliance with those requirements is ensured by appropriate corrective action taken by the provider of the system within an appropriate deadline set by the notified body.

For high-risk AI systems is important the presence of harmonized standards and common specifications. Providers can choose between various conformity assessment procedures to demonstrate compliance, depending on whether harmonized standards or common specifications are applied:

- **Standard request** is required by the Commission and serves to understand where improve AI systems' resource performance, such as reducing the consumption of energy and other resources during its lifecycle, and on the energy-efficient development of general-purpose AI models.

Then goals during the standardisation process by the participants are to promote investment and innovation in AI, including through increasing legal certainty, as well as the competitiveness and growth of the EU market, contributing to strengthening global cooperation on standardisation and considering already existing international standards. The Commission is empowered to adopt, implement acts establishing common specifications in case harmonised standards have not been accepted, are not given within the deadline set, or do not follow fundamental rights, compliances, or requirements.

The Act aims to ensure that high-risk AI systems are developed and used in a manner that upholds safety (conformity and compliance with cybersecurity), fundamental rights, and public trust. The provider shall have applied:

- The internal controls; or
 - The assessment of the quality management system and the assessment of the technical documentation, with the involvement of a notified body.
- **EU declaration of conformity** shall identify the high-risk AI system for which it has been drawn up. That shall contain all the information required to identify the European Union harmonisation legislation to which the declaration relates. The provider - who has drawn up the EU declaration of conformity - shall assume responsibility for compliance with the requirements and keep the EU declaration of conformity up to date as appropriate. Before placing it on the market or putting into service a high-risk AI system the provider or another individual representative shall register themselves and their system in the EU database; these will be registered at national level.

1.3 Transparency Obligations for Providers and Deployers of Certain AI System

When natural persons interact directly with AI systems, it is **responsibility of providers** to inform them that they are interacting with an AI system especially in contexts where they generate or manipulate content, such as synthetic audio, image, video, or text. This could be taken for granted when is obvious from the point of view of a natural person who is reasonably well-informed, observant, and circumspect, taking into account the circumstances and the context of use or when AI systems are authorised by law for a specific reason (to detect, prevent, investigate or prosecute criminal offences).

The effectiveness, interoperability, robustness, and reliability of these technical solutions are crucial, considering the limitations of content types, implementation costs, and the state of the art in relevant technical standards. Exceptions include AI systems used for standard editing assistance or those that do not significantly alter deployer-provided data or its semantics, and systems authorized for law enforcement purposes. Because there are systems that can recognize emotions and biometric categorization systems, the **deployers** are obligated to inform individuals exposed to these systems about their operation, subject to appropriate safeguards for the rights and freedoms of third parties, and in compliance with EU law.

AI system that generates a "deep fake", shall disclose that the content has been artificially generated or manipulated when it is created with the purpose of informing the public on matters of public interest. In addition, if the content forms part of an evidently artistic, creative, satirical, fictional analogous work or program, are allowed as long as the transparency obligations are appropriate and not hamper the display or enjoyment of the work. The obligations outlined on the AI technologies ensure that are utilized in a manner that is transparent, respectful of personal privacy, responsibly and consistent with fundamental rights.

The AI Office is tasked with encouraging the development of codes of practice at the EU level to aid in implementing these obligations effectively. The Commission has the power to adopt implementing acts to approve these codes or, if necessary, specify common rules for their implementation.

1.4 General-Purpose AI Models

General-purpose AI models are classified as having systemic risk based on their capabilities or impact. This classification is determined through technical evaluation, including indicators and benchmarks, or by a Commission decision following an alert from the scientific panel. A key quantitative threshold for classifying an AI model as having systemic risk is the use of computation exceeding 10^{25} floating-point operations (FLOPs) for its training. The Commission is empowered to adjust these thresholds and criteria to keep pace with technological advancements.

The provider shall notify the Commission that their general-purpose AI models meet systemic risk criteria within two weeks of learning that it will be satisfy. That notification shall include the information necessary to demonstrate that the relevant requirement has been met; without it the Commission may decide to designate it as a model with systemic risk. Under request by provider, the AI System could be reassessed by the Commission whether the request contains objective, detailed and new reasons that have arisen since the designation decision. Providers may request reassessment at the earliest six months after the designation decision.

All general-purpose AI models with systemic risk are contained in a list published and up to date, without prejudice to the need to observe and protect intellectual property rights and confidential business information or trade secrets in accordance with EU and national law. The information to share to the Commission shall include the technical documentation of the model, including its training and testing process and the results of its evaluation that should be up-to-date, constantly integrated and enable a good understanding of the capabilities and limitations. The information regarding also training and testing processes. In fact, the providers of general-purpose AI models with systemic risk shall:

- Perform model evaluation in accordance with standardised protocols and tools reflecting the state-of-the-art, including conducting and documenting adversarial testing of the model with a view to identifying and mitigating systemic risk;
- Assess and mitigate possible systemic risks and track their sources enabling identification of their origin;
- Keep track of document and report relevant information about serious incidents and possible corrective measures to address them;
- Ensure an adequate level of cybersecurity protection for the general-purpose AI model with systemic risk and the physical infrastructure of the model.

Providers of general-purpose AI models with systemic risk may rely on codes of practice to demonstrate that their compliance with the European harmonised standard; those who do not adhere to the approved code of practice shall demonstrate it with alternative adequate for achieve the approval by the Commission. The codes of practice shall include the following issues:

- To ensure that the information is kept up to date in the light of market and technological developments;
- An adequate level of detail for the summary about the content used for training;
- The identification of the type, the nature, and the sources of the systemic risks at EU level;
- The measures, procedures and modalities for the assessment and management of the systemic risks, taking into account their severity and probability of those risks may emerge and materialise along the AI value chain.

The **AI Office and the Board** shall aim to ensure that the codes of practice clearly set out their specific objectives and contain commitments or measures to ensure the achievement of those objectives considering the needs and the interests of all interested parties at European Union level.

1.5 Innovation

To encourage and facilitate the development of innovative AI technologies, it must be developed within a secure, compliant, and ethically grounded framework. To achieve this, the AI Act provides for the establishment of regulatory sandboxes for AI, tailored support measures for SMEs, including startups, and also includes specific exemptions for microenterprises. These measures enable the strengthening of the AI ecosystem while ensuring adherence to Regulation standards.

Regarding regulatory sandboxes, these refer to a controlled environment where supervised intermediaries and FinTech industry operators can test, for a limited period, technologically innovative products and services in the banking, financial, and insurance sectors[3]. The concept of regulatory sandboxes for AI is introduced as a controlled environment that fosters innovation by allowing the development, training, testing, and validation of AI systems under the supervision and guidance of competent authorities. Member States are required to establish at least one national **AI regulatory sandbox**, which could also be set up in collaboration with other Member States. Additionally, it is envisaged that additional regulatory AI sandboxes may be established at the regional or local level. The establishment of regulatory AI sandboxes aims to pursue the following objectives:

- Improve legal certainty to achieve regulatory compliance with the AI Act or other applicable EU and national laws;
- Support the sharing of best practices through cooperation with the authorities involved in the regulatory sandbox;
- Promote innovation and competitiveness and facilitate the development of an AI ecosystem;
- Contribute to evidence-based regulatory learning;
- Facilitate access to the EU market for AI systems, particularly when provided by SMEs, including startups.

According to the AI Act, national authorities must ensure the participation of data protection authorities and other authorities involved in the processing of personal data in the operation of regulatory sandboxes for AI. Providers in the sandbox remain liable for damages caused but will not be fined if they comply with directives and conditions. Additionally, regulatory AI sandboxes must be designed and implemented to facilitate cross-border cooperation among national competent authorities. Such authorities must submit annual reports to the AI Office and the Council, starting one year after the establishment of the sandbox and every subsequent year until its cessation, and a final report providing information on the progress and results of the implementation of such sandboxes.

In order to prevent fragmentation of regulatory AI sandboxes within the EU, the Commission can adopt implementing acts specifying detailed provisions for the creation, development, implementation, operation, and supervision. These implementing acts include common principles on issues, such as:

- Eligibility and selection criteria for participation in regulatory AI sandboxes;
- Procedures for application, participation, monitoring, exit, and termination of regulatory AI sandboxes, including sandbox plan and exit report;
- Terms and conditions applicable to participants.

The above-mentioned acts will ensure that regulatory AI sandboxes are accessible to any potential provider of an AI system that meets transparent and fair eligibility and selection criteria, and will also ensure broad and fair access, taking into account the demand for participation. Access to the sandboxes are free for SMEs, including startups, and potential providers will be facilitated through the learning outcomes of the sandboxes in fulfilling compliance assessment obligations required by law. Furthermore, the sandboxes facilitate the involvement of other actors in the AI ecosystem, enabling and facilitating cooperation with both the public and private sectors.

The AI Act provides that personal data collected legally for other purposes may be processed within a regulatory sandbox to develop, train, and test specific AI systems. The processed data must be

necessary to comply with specific regulatory requirements. Personal data must be kept in a separate and secure environment under the provider's control. The processing of personal data must not affect the decisions or rights of data subjects, and appropriate measures must be taken to protect data and ensure privacy. AI systems must be developed to safeguard a substantial public interest by a public authority in areas such as public safety, healthcare, environmental protection, energy sustainability, transportation security, and public administration efficiency.

1.5.1 Testing of High-Risk AI System in Real World Conditions Outside AI Regulatory Sandboxes

High-risk AI systems may undergo testing outside of regulatory sandboxes. Testing of high-risk AI systems in real-world environments outside of defined regulatory areas for AI may be conducted by providers or potential providers of such systems listed in a reference document, following a set of specific guidelines and procedures. These guidelines include preparing a detailed testing plan, obtaining approval from a competent authority, and ensuring the protection of individuals involved in the test. Before participating in such tests, obtaining informed consent from the individuals involved is necessary. These individuals must be fully informed about the nature and objectives of the test, the methods used, the expected duration, as well as their rights during the test, including the right to withdraw without consequences. They must also be informed about how they can request modifications or reject decisions made by the AI system. Lastly, it is important that consent is documented, and a copy is provided to the participants or their legal representatives.

Furthermore, the AI Act establishes specific measures that suppliers and users, especially SMEs and startups, must adhere to. Indeed, Member States are required to actively support SMEs and startups within the framework of AI regulation, ensuring their priority access to regulatory AI sandboxes while respecting established requirements. Additionally, Member States must organize awareness and training activities tailored to the needs of these companies, involving local public authorities if necessary, and maintain dedicated communication channels and promote SME participation in standard development.

Meanwhile, the AI Office plays a key role in ensuring regulatory compliance by providing standardized templates, informative platforms, and promoting best practices in the procurement processes of AI systems. Microenterprises may comply with certain elements of the required quality management system in a simplified manner, provided they do not have partners or linked enterprises that would disqualify them from such simplifications. To this end, the AI Act stipulates that guidelines will be developed to assist these microenterprises in understanding and implementing these simplified compliance measures.

1.6 Governances

The **European Artificial Intelligence Board** is hereby established and represents a significant initiative aimed at fostering closer collaboration among the Member States in the field of AI. This body will consist of one representative from each member country, with the European Data Protection Supervisor participating as an observer. Despite lacking voting rights, the AI Office attends Council discussions. Operational procedures of the Council are outlined, with its presidency assigned to one of the representatives from the Member States, while the AI Office handle meeting organization and provide necessary administrative support. Member countries select their representatives for a three-year term, with the option for renewal once. These representatives must possess the necessary expertise to effectively carry out the tasks assigned by the Council.

The AI Office aims to ensure that the Council can carry out its activities effectively and impartially, promoting adequate governance in the AI sector. The Council, on the other hand, consult and assist the Commission and the Member States to ensure consistent application of the AI Act and it assists national authorities and the Commission in skill development, provides guidance on drafting guidance documents, and offers opinions on international AI-related matters and qualified alerts regarding general-purpose AI models. Among its functions, the Council coordinates the involved national authorities, shares knowledge and best practices, provides guidance on AI Act implementation, harmonizes administrative practices, issues recommendations on various matters, supports the promotion of AI literacy, and contributes to effective international cooperation. Finally,

it considers the opinions of the Member States on the implementation of the regulation and the monitoring of AI systems.

Furthermore, there is a consultative forum that provides technical expertise and advice to the Council and the Commission, thereby supporting their tasks under this regulation. This consultative forum is composed in a diverse and balanced manner, including representatives from industry, start-ups, SMEs, civil society, and academia. selected by the Commission from among those with recognized expertise in the field of AI. Members will serve a two-year term, extendable up to four years. From certain organizations such as the Fundamental Rights Agency, ENISA, the European Committee for Standardization (CEN), the European Committee for Electrotechnical Standardization (CENELEC), and the European Telecommunications Standards Institute (ETSI) the members shall be permanent. Member States may engage experts from the scientific panel to support their implementation activities of the regulation, potentially subject to fees. To this end, the Commission takes care of facilitating the access of such experts, which must occur promptly, and will ensure efficient organization of the support provided by both the experts and the European Union's AI support, aiming to maximize added value. Additionally, Member States may request assistance from experts of the scientific panel to implement the regulation. They may be subject to fees for the support received, which will be determined in the implementing act taking into account the regulation's objectives, cost-effectiveness, and fair access to experts.

Each Member State is required to establish or designate notifying authorities and market surveillance authorities for the purposes of the regulation. These authorities must act independently, impartially, and without bias, and must be equipped with adequate resources and expertise in AI technologies and related fields. Member States must also designate a single point of contact for the regulation and submit a report to the Commission every two years on the resources and adequacy of national competent authorities. Additionally, they may engage experts from the scientific panel to support their enforcement activities under the regulation, whose access is facilitated by the Commission, ensuring efficient organization of support activities. Furthermore, these experts can provide assistance for the implementation of the regulation and may charge fees, which will be determined in the implementing act, for the support provided to the Member States. These authorities must act independently, impartially, and without bias, equipped with adequate resources and expertise in AI technologies and related fields. Member States must also designate a single point of contact for the regulation and submit a biennial report to the Commission on the resources and adequacy of national competent authorities.

1.7 EU Database for High-Risk AI System

The database serves as a critical component of the EU's regulatory framework for AI, designed to enhance transparency, accountability, and oversight of high-risk AI systems. The European Commission, in collaboration with Member States, is tasked with setting up and maintaining a properly **EU database** with functional specifications determined in consultation with relevant experts and the **European Artificial Intelligence Board** for updates, ensuring that it meets the evolving needs of regulatory oversight.

The database is designed to include specific information sets which providers or their authorized representatives must enter. This structured approach ensures comprehensive documentation of high-risk AI systems, including their development, deployment, and compliance status. The information contained in the database shall be accessible and publicly available in a user-friendly manner easily navigable and machine-readable.

Nevertheless, there is information available exclusively to market surveillance authorities and the Commission unless consent for public access is provided by the provider or prospective provider. This tiered access safeguards sensitive information while promoting transparency where appropriate. The inclusion of personal data in the database is limited to what is necessary for fulfilling the Regulation's objectives, focusing on the names and contact details of individuals responsible for system registration with legal authority to represent providers or deployers. This measure balances the need for accountability with privacy and data protection principles.

The Commission acts as the controller of the EU database, providing technical and administrative support to providers, prospective providers, and deployers to facilitate compliance. The database's design and operation adhere to applicable accessibility requirements, ensuring it is user-friendly

and accessible to all stakeholders. Through careful management of personal data and a commitment to accessibility, the EU database embodies the Regulation's principles of transparency, accountability, and privacy.

1.8 Post-Marketing Monitoring, Information Sharing, Market Surveillance

1.8.1 Post-Market Monitoring

Providers are mandated to establish a post-market monitoring system, which is proportional to the AI technology's nature and associated risks. This system should actively collect, document, and analyze relevant data to evaluate the AI systems' ongoing compliance with requested requirements. The data can come from deployers or other sources and must include performance metrics across the AI systems' lifecycle. A post-market monitoring plan, part of the technical documentation, guides this process. The Commission is tasked with developing a template for this plan, ensuring consistency and thoroughness in monitoring practices.

1.8.2 Sharing of Information on Serious Incidents

Providers of high-risk AI systems placed on the EU market shall report any serious incident to the market surveillance authorities of the Member States where that incident occurred. The timeframe for reporting depends on the incident's severity, with a general guideline of no later than 15 days after recognizing a causal link between the AI system and the incident. For particularly grave incidents, such as those leading to death, reports must be made within two days. Providers are required to investigate serious incidents promptly and cooperate with competent authorities, ensuring that any necessary corrective action is taken. National competent authorities shall immediately notify the Commission of any serious incident, whether they have acted on it.

1.8.3 Enforcement

Under Regulation (EU) 2019/1020[15], the AI Regulation shall be aligned with the broader framework of market surveillance, defining the scope of market surveillance activities, and clarifying the roles of various national authorities in overseeing high-risk AI systems. The procedures shall not apply to AI systems when exist already relevant sectoral procedures ensuring an equivalent level of protection and having the same objective. Guaranteeing full access to the documentation and the training, as well as validation and testing data set used for the development of high-risk AI systems (i.e., application programming interfaces ("API")) or other relevant technical means and tools enabling remote access to have as purpose to ensure the compliance with the Regulation. This includes overseeing testing within AI regulatory sandboxes and verifying adherence conditions. Authorities have the power to modify, suspend, or terminate testing if necessary to protect public safety and compliance with the regulation.

Respect the confidentiality of information and data obtained in carrying out their tasks and activities in such a manner as to protect both the intellectual property rights, confidential business information or trade secrets of a natural or legal person. Put in place adequate and effective cybersecurity measures and shall delete the data collected as soon as it is no longer needed for the purpose for which it was obtained, in accordance with applicable EU or national law.

1.8.4 Remedies

For to allow any individual or entity to submit complaints to authorities if they believe there has been a breach of the Regulation's provisions, are available two remedies toward high-risk AI systems, more specifically:

- **Right to Lodge** a Complaint with a Market Surveillance Authority when it is believed there has been a breach of the Regulation's provisions: such complaints shall be taken into account for the purpose of conducting market surveillance activities and shall be handled in line with the dedicated procedures established therefor by the market surveillance authorities.

- **Right to Explanation** of Individual Decision-making; it allows to affected persons have the right to obtain clear and meaningful explanations from deployers about the role and impact of high-risk AI systems in individual decision-making processes, especially when these decisions significantly affect their rights, health, or safety.

1.8.5 Supervision, Investigation, Enforcement and Monitoring in Respect of Providers of General-purpose AI Models

The Commission has the power to supervise and enforce and, for the implementation of these tasks, shall entrust the activity to the AI Office which may take the necessary actions to monitor the effective implementation and compliance with the Regulation by providers of general-purpose AI models, including their adherence to approved codes of practice. The market surveillance authorities may request the Commission to exercise specific powers when and proportionally to assist with the fulfilment of their tasks.

Providers shall have the right to lodge a complaint alleging an infringement of the Regulation and, when the alerts come from the scientific panel, defined qualified alert, is the Commission that assesses the matter. For both above situations, the Commission may request the provider of the general-purpose AI model concerned to provide the documentation drawn up and any additional information that is necessary for the purpose of assessing compliance of the provider or the general-purpose AI model.

These actions are critical in ensuring the ongoing compliance of AI systems with the Regulation's standards, safeguarding public safety, and protecting fundamental rights.

1.9 Codes of Conduct and Guidelines

Regarding the role and formulation of codes of conduct and guidelines, their aim is to promote the voluntary application of specific requirements to AI systems beyond those classified as high-risk. Referring the **codes of conduct**, the AI Act establishes that the AI Office, alongside Member States, have the task of encouraging and facilitating their creation. These codes, aiming to broaden the scope of requirements to encompass AI systems not classified as high-risk, are designed to incorporate industry-specific technical solutions and best practices. This initiative reflects a wider aspiration to promote the ethical development of AI beyond the confines of regulatory mandates. The AI Office and Member States are tasked with fostering the creation of voluntary codes for the use of AI systems, characterized by specific requirements aligned with clear objectives and measurable performance indicators, including EU ethical guidelines, environmental impact assessment and mitigation, promotion of AI literacy, inclusive system design, and prevention of adverse effects on vulnerable groups, including individuals with disabilities, and promotion of gender equality. Additionally, such codes may be developed by individual providers or users, their representative organizations, or through collaborative efforts involving stakeholders from civil society and academia. As for their intended purposes, these are developed to address the unique needs and challenges encountered by SMEs, including startups. This enables smaller entities to readily adopt and adhere to elevated ethical standards and practices, thereby promoting widespread commitment to responsible AI across the ecosystem.

Regarding **guidelines**, the Commission is responsible for drafting guidelines to facilitate the practical implementation of the Regulation, covering a wide range of areas from the application of specific requirements and obligations to the management of prohibited practices and transparency obligations. These guidelines are designed to clarify and facilitate compliance, ensuring that stakeholders across the AI landscape can effectively navigate the regulatory framework. The Commission, according to AI Act, has produced guidelines for the practical implementation of the AI Regulation. In drafting these guidelines, the Commission must consider the specific needs of SMEs, including startups, local public authorities, sectors most likely to be influenced by the Regulation, and additionally, the state of the art in AI, relevant harmonized standards and common specifications, as well as provisions on EU harmonization. Recognizing the rapid evolution of AI technology and its applications, the Commission is tasked with updating these guidelines as necessary, whether at the request of Member States, the AI Office, or on its own initiative. This adaptive approach ensures that the guidance provided remains relevant and responsive to

technological advancements and emerging best practices in the field.

1.10 Delegation of Power and Committee Procedure

The AI Act regulates delegated acts, outlining the procedural aspects related to the committee assisting the Commission.

1.10.1 Delegated Acts

The European Commission is empowered to adopt delegated acts, allowing it to amend non-essential elements of the Regulation or integrate it, subject to the procedures outlined in this chapter.

The delegation of powers is initially granted for a period of five years from the entry into force of the Regulation, with automatic renewal for identical periods unless there is explicit opposition from the European Parliament or the Council.

The Commission also has the power to adopt specific delegated acts in various articles for a period of five years from the date of entry into force of the regulation. Additionally, it must submit a report on the delegation of powers no later than nine months before the end of the five-year period. The renewal of the delegation will be automatic unless the European Parliament or the Council oppose it within three months before the end of each period. The delegation of powers outlined in the AI Act can be revoked by the European Parliament or the Council at any time and will take effect from the day following its publication in the Official Journal of the European Union.

Before adopting a delegated act, the Commission is required to consult experts designated by each Member State, respecting the principles of best legislation, and ensures that delegated acts reflect the experience and perspectives of the Member States, promoting a collaborative approach to regulation. After adopting a delegated act, the Commission must notify both the European Parliament and the Council simultaneously, maintaining a transparent and inclusive legislative process.

1.10.2 Committee Procedure

The committee provides its opinion on acts that must be adopted on the proposal of the Commission:

- If the committee expresses a positive opinion, the Commission adopts the draft implementing act;
- If the committee expresses a negative opinion, the Commission does not adopt the draft implementing act. However, the President of the Commission can submit a modified version to the committee within a month or to the appeal committee for a new decision;
- If no opinion is expressed, the Commission adopts the draft implementing act. If the Commission does not adopt the draft implementing act, the President can submit a modified version to the committee.

The Commission does not adopt the draft implementing act if it concerns certain sectors (taxation, financial services, protection of human, animal, or plant health or safety, multilateral definitive safeguard measures), if the basic act provides for it, or if a simple majority of the committee members opposes it. The president may submit a modified version to the same committee within two months or to the appeal committee within one month for a new deliberation, based on the aforementioned cases of non-adoption. The Commission consults with the Member States and informs the committee of the results. The appeal committee convenes to provide its opinion on the draft implementing act.

1.11 Penalties

The AI Act outlines the framework for imposing **sanctions** on operators who violate the provisions established therein. These aim to enforce the provisions of the AI Act and ensure compliance, thus protecting the public interest, fundamental rights, and safety standards in the context of AI development and implementation.

In accordance with the AI Regulation, Member States must establish sanction rules and implement enforcement measures for operator violations. These rules may include **warnings** and **non-monetary**

measures and must be effectively implemented, taking into account the Commission's guidelines. Sanctions must be effective, proportionate, and dissuasive, also considering the interests of SMEs and startups. Member States must notify the Commission of the sanction rules and other enforcement measures, promptly informing it of any subsequent changes. Operators that do not comply with the prohibition of certain AI practices (see chapter "Prohibited AI practices") are subject to **administrative fines** of up to €35 million or up to 7% of the total worldwide annual turnover for the preceding financial year, whichever is higher. Additionally, non-compliance with obligations related to providers, authorized representatives, importers, distributors, deployers, notified bodies, and transparency obligations can result in fines of up to €15 million or up to 3% of the total worldwide annual turnover for the preceding financial year, whichever is higher. Providing incorrect, incomplete, or misleading information to notified bodies or national competent authorities in response to a request is subject to administrative fines of up to €7.500.000 or up to 1% of the company's total worldwide annual turnover for the preceding financial year, whichever is higher. For SMEs, including startups, each fine is capped at lower percentages or amounts as specified in the Regulation, ensuring penalties do not disproportionately impact smaller businesses.

When assessing the imposition and quantification of an administrative sanction in a specific case, it is essential to consider all relevant factors of the situation. These include the nature, gravity, and duration of the infringement and its consequences, taking into account the purpose of the AI system and the extent of people's involvement and the damages incurred. It is also important to ascertain whether other market surveillance authorities have already penalized the same operator for the same violation or if sanctions have been imposed for breaches of other laws, in addition to evaluating the size, annual turnover, and market share of the operator. Other significant factors include the financial benefits gained or losses avoided as a result of the infringement, the level of cooperation with authorities to resolve the violation, the degree of responsibility, and the actions taken by the operator to mitigate the harm suffered by those involved. Furthermore, it is crucial to determine whether the infringement was intentional or due to negligence.

Member States are tasked with establishing rules on how administrative fines may be imposed on public authorities and bodies within their jurisdiction. Depending on the legal system of the Member States, the provisions regarding administrative sanctions can be implemented in various ways: penalties may be imposed by competent national courts or by other bodies, in accordance with national laws. However, the application of such provisions must ensure an equivalent effect across all relevant Member States. The market surveillance authority, in exercising its powers, must adhere to appropriate procedural safeguards in line with both EU and national law, ensuring the possibility of effective judicial remedies and adherence to the principle of due process. Finally, on an annual basis, Member States must report to the Commission the administrative fines issued during that year, as well as any related legal or judicial proceedings.

1.12 Entry into Force and Review

Regarding AI systems that have already been placed on the market or put into service before certain deadlines specified in the Act, AI Act outlines a timeline for bringing these systems into compliance with the Act's requirements, providing a transition period for operators to adjust to the new regulatory environment. The AI Regulation shall enter into force on the twentieth day following that of its publication in the Official Journal of the European Union. It shall apply from 24 months from the date of entry into force of this Regulation. However:

- Provisions regarding **Prohibited** AI system shall apply from **6 months** from the date of entry into force of this Regulation (mid-December 2024);
- Provisions and rules regarding **General Purpose** AI shall apply from **12 months** from the date of entry into force of this Regulation (mid-July 2025);
- Provisions and rules regarding **High risk** AI system shall apply from **24 months** from the date of entry into force of this Regulation (mid-July 2026);
- Provisions and rules regarding for **AI related to Harmonized Products** shall apply from **36 months** from the date of entry into force of this Regulation (mid-July 2027).

This provision reflects the Act's forward-looking approach while acknowledging the need for a gradual transition for existing AI systems.

The Commission is tasked with regularly evaluating and reviewing the Act to assess the need for amendments, including updates to the list of high-risk AI systems and prohibited AI practices (by four years from the date of entry into force of this Regulation and every four years thereafter). The evaluation process will consider the Act's impact on market entry for new undertakings, particularly SMEs, the adequacy of resources for national competent authorities, the application of penalties, and the development of harmonized standards. This ongoing evaluation ensures that the Act remains relevant and effective in the face of technological advancements and market developments. The AI Office is expected to develop a methodology for evaluating risk levels and including new systems in the lists of high-risk AI systems, prohibited practices, and systems requiring additional transparency measures. This support is crucial for maintaining a dynamic and responsive regulatory approach to AI.

The Regulation will enter into force on the twentieth day following its publication in the Official Journal of the European Union and will be applicable from 24 months after its entry into force. However, certain chapters and provisions have different application timelines to ensure a phased implementation of the Act; (i) Provision regarding prohibited AI system shall apply from six months from the date of entry into force of this Regulation and (ii) other provision regarding high-risk AI system shall apply from 12 months from the date of entry into force of this Regulation. This staged approach facilitates a smooth transition for stakeholders to adapt to the new regulatory requirements.

2. AI Regulation in Other Countries and the Actions

Regarding other **extra-Europe countries**[19][26], on October 30th 2023, **Unites States** President Biden signed an *"Executive Order (EO) on Safe, Secure, and Trustworthy Artificial Intelligence"*[28]. This EO builds on previously published guidance documents such as the AI *"Bill of Rights"*[27] and the National Institute for Standards in Technology (NIST) AI Risk Management Framework[22]. The order defines AI broadly and lays out eight principles and priorities for the use of AI:

- AI must be safe and secure;
- The US should promote responsible innovation, competition and collaboration in AI development;
- The responsible use and development of AI must come with a commitment to supporting American workers;
- AI policies must be consistent with the Biden Administration's dedication to improving equity and civil rights;
- Interests of Americans who increasingly use, interact with, or purchase AI and AI-enabled products in their daily lives must be protected;
- Americans' privacy and civil liberties must be protected;
- The risks from the Federal Government's own use of AI must be adequately managed and the government's capacity to;
- The US Federal Government should lead the way and engage with international partners to develop a framework to manage AI risks and unlock potentials.

China has implemented interim measures to address AI-related risks and impose compliance obligations on entities engaged in AI-related business. Provisions only apply to public usages of gen-AI⁴ by both domestic and foreign individuals and entities involved in AI services in China. A

⁴Generative artificial intelligence (gen-AI) is artificial intelligence capable of generating text, images, videos, or other data using generative models, often in response to prompts. Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics.

draft version of the interim measures was released for comments in April 2023[6], followed by a substantially revised version dated July 2023 [5] and taking effect on August 2023. Similar to the EU AI Act, the measures do not apply to gen-AI technologies used for research purposes and not deployed to the market. Encouragement for gen-AI development was also added. Key points of the July 2023 *"Interim Measures for the Management of Generative Artificial Intelligence Services"* are:

- Gen-AI services providers with public opinion properties or with the capacity for social mobilization shall carry out security assessments in accordance with relevant state provisions.
- Obligor must comply with laws, social morality, and ethics, and avoid manipulating information or public opinion. During algorithm design, selection of data, model generation and training, and provision of services, effective measures should be employed to prevent biases that result in discrimination. Providers shall fulfill confidentiality obligations towards information input by users and users' usage records in accordance with existing laws.
- Content is prohibited if it is against *"Core Socialist Values"* or if it otherwise endangers national security. Providers are responsible for the legality of content generated and diffused, and for prompt interruption of services and rectification of unlawful content.
- Intellectual property rights should be protected and advantages in algorithms, data, platform and the like must not be used for monopolies or to carry out unfair competition.
- Penalties for non-compliance include warnings, ordering rectifications and corrections, suspension of services, as well as civil and criminal prosecution when relevant.

In the UK, several authorities are involved in the regulation of AI, and the regulatory approach is characterized by guidelines, best practices and principles. Examples of key initiatives include *"Defence Artificial Intelligence Strategy"*[20], *"Guidance on AI and Data Protection"*[29] and the UK's NCSC's *"Guidelines for Secure AI System Development"*[21]. In March 2023, the UK Government issued a white paper for *"A Pro-Innovation Approach to AI Regulation"*[25]. This paper stresses a regulatory attitude based on specific areas of application of AI rather than specific technologies. It suggests that potential future regulation shall focus on principles such as safety, transparency, fairness, accountability, and contestability. The paper excludes immediate regulatory action, delineating instead plans to invest in AI research and development, and to collaborate with international partners to influence global AI governance. The UK Government is currently working on a *"Code of Practice"* on Copyright Issues and AI[18].

AI's impacts have a global component that calls for international cooperation. In addition to strategic considerations, the need for cooperation emerges from externalities inherent to the technology that may spill over national borders, or regulation themselves. The cooperation may be complicated by contrasting objectives and definitions of social welfare. Below it is reported some initiatives relative to international cooperation and multilateral actions relative to AI[19]:

- **OECD "AI Principles"**[24] - In May 2019, the OECD adopted a set of *"AI Principles"* to foster innovative and trustworthy AI that respects human rights and democratic values. The five principles are: i) inclusive growth, sustainable development and well-being, ii) human-centered values and fairness, iii) transparency and explainability, iv) robustness, security and safety, and v) accountability. The principles have since been endorsed by 46 countries worldwide and have been embedded in several national and multinational initiatives.
- **NATO AI strategy** - In its October 2021 meeting, the NATO Allied Defence Ministers formally adopted an AI strategy for defense and national security, committing to the cooperation and collaboration necessary for its implementation. Signatories committed to *"Principles of Responsible Use"*[23] for the development and deployment of AI. The six principles are: (i) Lawfulness; (ii) Responsibility and Accountability; (iii) Explainability and Traceability; (iv) Reliability; (v) Governability; and (vi) Bias Mitigation.
- **World Economic Forum** - In June 2023, The World Economic Forum (WEF) launched the *"AI Governance Alliance"*[31] to unite industry leaders, governments, academic institutions and civil society organizations around the goal of meaningful AI governance. Ultimately, AI's

opportunities and challenges, including governance and regulation, were at the center of the 2024 Davos meetings.

- **UN initiatives** - In December 2023, The UN AI Advisory Body released the Interim Report *"Governing AI for Humanity"*[30]. The document calls for a closer alignment between international norms and AI's developed and deployment. The AI Advisory Body recommends five guiding principles: i) AI should be governed inclusively, by and for the benefit of all; ii) AI must be governed in the public interest; iii) AI governance should be built in step with data governance and the promotion of data commons; iv). AI governance must be universal, networked and rooted in adaptive multi-stakeholder collaboration; v). AI governance should be anchored in the UN Charter, International Human Rights Law, and other agreed international commitments such as the Sustainable Development Goals.
- **G7: "Hiroshima Process International Code of Conduct for Advanced AI Systems"** - On October 30, 2023, G7 leaders agreed to the *"Hiroshima Process International Code of Conduct for Advanced AI Systems"*. The code provides *"voluntary guidance for actions by organizations developing the most advanced AI systems"*[17]. The document lays out guidance principles encouraging: i) risk mitigation in all parts of the AI process; ii) increased transparency during development reporting systems' capabilities and domains of use; iii) sharing of information and reporting of incidents in development; iv) developing governance, v) increasing security controls and advance the development of international standards; vi) deploying reliable provenance mechanisms such as watermarking; vii) prioritizing research to increase AI's safety and on applications that would help sustainable development goals; viii) implementing data input measures and protection for personal data and intellectual property.
- **The first global "AI Safety Summit"** - On November 1-2, 2023 the British government hosted the first global *"AI Safety Summit"*, gathered representatives from 28 national governments. The initiative discussed how to approach and regulate AI technologies. On this occasion, the participating national governments and the EU signed the *"Bletchley Declaration"*[1], resolving to further national, multilateral and bilateral action to promote on AI safety research and establish risk-based policy across the respective countries.
- **Guidelines for Secure AI System Development** - In November 2023, the UK National Cyber Security Centre (NCSC), the US Cybersecurity and Infrastructure Security Agency (CISA), and several other national cyber security agencies across AEs and EMDEs released a set of Guidelines[21]. These guidelines aim to: a secure design, secure development, secure deployment, and secure operation and maintenance. The report includes suggestions about mitigations to related risks in the view that providers of AI components should take responsibility for the security outcomes of users further down the supply chain.
- **The Global Partnership on AI and the Ministerial Declaration** - The goal of the Global Partnership on AI is to guide the responsible development and use of AI. On occasion of the December 2023 general purpose AI summit in Delhi, Ministers of the 29 member countries signed a joint declaration reaffirming their commitment to promote responsible and trustworthy AI, and their dedication to jointly develop regulations, policies, and standards to uphold general-purpose AI's principles[16]. The Ministers also embraced the notion of *"collaborative AI"*, which involves supporting and promoting equitable access to critical resources for AI research and innovation, such as AI computing, high-quality diverse datasets, algorithms, software.

3. Final Provisions

The evolution of AI has been marked by rapid advancements in technology that have fundamentally transformed various sectors, including healthcare, transportation, finance, and more. Globally, AI's growth has prompted significant economic, social, and ethical implications. In the European Union, the implications of AI's evolution have been met with a proactive approach aimed at harnessing its benefits while addressing potential risks. The EU has recognized the need for a regulatory

framework that ensures AI's development and use align with core values and fundamental rights, such as privacy, dignity, equality, and non-discrimination. The AI Act aims to establish harmonized rules for the development, marketing, and use of AI systems within the European Union. It focuses on ensuring AI systems are human-centric, trustworthy, and uphold fundamental rights, including privacy, non-discrimination, and consumer protection. Overall, the objectives of the AI Act reflect the European Union's ambition to lead in the development of ethical, secure, and cutting-edge AI technologies. The Act's successful implementation will require ongoing dialogue between regulators, industry stakeholders, and society to adapt to the evolving landscape of AI technologies.

While the Act aims to protect citizens' rights and safety, it also seeks to foster innovation and competitiveness in the AI sector, thanks to a continuous balance between regulation and encouragement of technological development, ensuring the EU remains a global leader in trustworthy AI.

However AI Act faces several challenges, these concerns primarily revolve around balancing innovation with regulation, addressing practical challenges in classification and compliance, and understanding the potential impacts on small and medium-sized enterprises (SMEs).

- One of the critical challenges of the AI Act is ensuring that the regulatory framework supports and does not stifle innovation.
- The Act's risk-based approach to AI regulation necessitates the classification of AI systems according to the level of risk they pose. This classification process could be complex and subject to interpretation, raising concerns about consistency and clarity in its application. Moreover, the requirement for high-risk AI systems to comply with stringent regulations poses practical challenges for developers and deployers in terms of technical documentation, data governance, and ensuring human oversight.
- SMEs play a crucial role in the EU's economy and are often at the forefront of innovation in AI. However, the regulatory obligations imposed by the AI Act could disproportionately affect these smaller entities. The costs associated with ensuring compliance, such as conducting risk assessments, maintaining extensive documentation, and implementing data governance and human oversight mechanisms, could be particularly challenging for SMEs with limited resources, hindering their ability to compete with larger companies and stifling innovation within the sector.

In response to these challenges and criticisms, ongoing dialogue between policymakers, industry stakeholders, researchers, and civil society is crucial. This dialogue can help refine the regulatory approach to ensure it remains effective, proportionate, and conducive to innovation. Moreover, the dynamic nature of AI technology necessitates regulatory frameworks that are flexible and adaptable, allowing for updates and revisions as the technology and its applications evolve. Given the global nature of AI development and deployment, international cooperation is crucial for establishing consistent standards and practices for AI governance. Collaborative efforts can help address cross-border challenges, such as data privacy, security, and the ethical use of AI technologies. The EU can play a leading role in fostering international dialogue, sharing best practices, and contributing to the development of global norms for AI.

Furthermore, in the light of *Digital Operational Resilience Act (DORA)*[14] on the ICT sector, financial institutions should start to consider how DORA's imposed requirements interact with the obligations stemming from the AI Act. Indeed, DORA aims to ensure the ICT resilience of the EU financial sector and represents both a challenge and an opportunity to support, reboot and strengthen cyber resilience. The regulation addresses information security and ICT third party management for all regulated financial entities, introducing new requirements where gaps exist: (i) ICT risk management, (ii) ICT incident reporting, (iii) digital operational resilience, (iv) third-party ICT risk management and (v) information sharing agreements. Traditionally strong reliance of financial institutions on third-party ICT services will become even more prominent in the context of AI. Due to the lack of internal capabilities for the development of AI solutions, outsourcing to ICT service providers[7] is expected to increase, as will the security issues and challenges to the governance framework of institutions, particularly internal controls, data management and data protection.

Following the formal approval of the text of the AI Act, organizations and financial institutions can proactively begin preparing for compliance. The first step is to ensure the right people in the

organization start preparing for these upcoming regulatory requirements as soon as possible. Early engagement give more time to understand the requirements and their impact across the AI lifecycle. The AI Act identifies various roles, including legal, privacy, data science, risk management and procurement professionals. A multidisciplinary task force responsible for compliance with the AI Act should cover this full range of expertise. The second step is to comprehensively understand AI systems developed or used in the organization and categorize them based on the risk levels defined in the AI Act, also with a proper inventory of this models. If any of AI systems fall into the minimal, high, or unacceptable risk category, you may be required to make significant changes to processes and operations before 2026 or sooner for AI systems with unacceptable risk. It is crucial to have a clear plan of what needs to be done as quickly as possible to manage the necessary organizational transformation and ensure timely compliance with the new legal framework when it comes into effect and transparency to stakeholders. When an AI system is not compliant with regulations, thorough gap analyzes should be conducted to identify areas of non-compliance and develop an action plan to address these gaps.

Regarding the financial institutions activities, AI systems used to evaluate the credit score or credit-worthiness of natural persons should be classified as high-risk AI systems, since they determine those persons' access to financial resources. AI systems used for those purposes may lead to discrimination between persons or groups and may perpetuate historical patterns of discrimination, such as that based on racial or ethnic origins, gender, disabilities, age or sexual orientation, or may create new forms of discriminatory impacts. However, internal AI systems provided for by Union law for the purpose of detecting fraud in the offering of financial services and for prudential purposes to calculate credit institutions' and insurance undertakings' capital requirements should not be considered to be high-risk under this Regulation. Finally, AI systems intended to be used for risk assessment and pricing in relation to natural persons for health and life insurance can also have a significant impact on persons' livelihood and if not duly designed, developed and used, can infringe their fundamental rights and can lead to serious consequences for people's life and health, including financial exclusion and discrimination.

In conclusion, the AI Act represents a significant stride toward creating a harmonized and ethical framework for AI technologies. The Act's emphasis on fundamental rights and ethical considerations sets a global standard for AI governance, encouraging other regions to consider similar comprehensive and principled approaches. By balancing the promotion of innovation with the protection of individual rights, the EU aims to foster an AI ecosystem that is both dynamic and responsible. The AI Act is a landmark piece of legislation with far-reaching implications for the future of AI in the EU and beyond. It underscores the EU's commitment to leading the way in ethical AI governance, setting a benchmark for the world and ensuring that AI serves the public good while respecting human rights and democratic values. 

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